KCE PATHWAYS · GRADE 3 · UNIT 3: MARKET DAYS · WEEK 14 · SCIENCE

Engineering Challenge: Design It, Test It, Fix It

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Before the Lesson: Be an Engineer

Write or draw what you already know. Guesses are welcome!

1. What do you think an engineer does?

2. **Predict:** Which holds more pennies, a flat sheet of paper or a folded one? Why?

3. Draw a bridge you could make from one sheet of paper.

4. What is one problem at home you wish you could solve?

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

After the Lesson: My Bridge Test

Build and test your bridge with a grown-up. Then show what you learned.

1. Write the six design steps in order.

2. Draw your best paper bridge. Label the folds and write how many pennies it held.

3. **Evidence:** Which shape was strongest, in your test or the lab table? How do you know?

4. Choose Lonnie Johnson, Mary Anderson, Margaret Knight, or William Kamkwamba. How did this person test and fix a design?

★ **Challenge:** Add a new constraint: use only half a sheet of paper. Can your bridge still hold 20 pennies? Test it and write what you changed.

Looking Ahead: Next week is the **Unit 3 Checkpoint**, the last lesson of the course! Review solids, liquids, and gases, pushes and pulls, light and sound, and the design process. Bring one question you still have.